

SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Engineering Conformance Specification

Scientific contract version: v0.1

Document revision: r0.5

Status: Unassessed conformance procedure

Canonical science source: SCI-MAP v0.1 shared authority

Target implementation: Not selected in this document

No independent science. Every normative scientific clause and every prediction printed here is rendered from SCI-MAP-v0.1_SHARED_AUTHORITY_r0.1.tex. This document adds only evidence-recording, execution, traceability, and pass/fail procedure. If an engineering observation appears to require changing science, record a deviation and return it to scientific-owner review; do not edit the expected result in this view.

Contents

1	Purpose and conformance target	2
2	Evidence record	2
3	Claim-layer gates	3
4	Execution order and atomicity protocol	3
5	Canonical requirement conformance clauses	5
6	Prediction execution matrix	18
7	Mechanical completeness checks	22
8	Required handoff contents	22

1 Purpose and conformance target

This specification tells a reviewer how to decide whether one exact candidate conforms to SCI-MAP v0.1. It does not select a candidate, inspect an implementation, execute validation, or assert a result. Before work begins, the reviewer freezes:

- repository and immutable candidate identity;
- build profile, compiler/runtime identities, configuration, and enabled parallel modes;
- exact input fixtures and independent expected-value generators;
- numerical comparison policy and simultaneous statistical error policy;
- required product list, representation paths, log policy, and completion criteria; and
- the shared-authority digest, contract version, this document revision, and owner-decision ledger revision.

Evidence from a different candidate, configuration, or science-authority digest is not silently pooled. Each rerun keeps its own identity and reason. Historical operational acceptance does not substitute for conformance evidence.

2 Evidence record

Each requirement record contains at least:

Field	Required content
Requirement	Stable SCI-MAP-REQ-NNN and shared-authority digest.
Candidate	Exact source/build identity and environment profile.
Observable	Public product, typed state, API boundary, metadata, diagnostic, or failure transition under test.
Fixture	Immutable input identity and independent expected-value calculation.
Comparison	Exact predicate or preregistered numerical/statistical bound.
Result	Pass, fail, blocked by named owner decision/dependency, or not applicable with owner-approved rationale. Skipped required evidence is not a pass.
Artifacts	Machine-readable outputs, logs, traces, plots/tables where needed, and digest-addressed provenance.
Reviewer	Independent reviewer identity, date, and deviation disposition.

Failures are retained as evidence. A tolerance, fixture, expected result, or applicability decision is not changed after viewing a failure without opening a new preregistered record.

3 Claim-layer gates

Passing one layer grants no stronger layer automatically.

Layer	Minimum engineering record	Prohibited inference
Algebraic contract	Owner-frozen shared authority and internally consistent independent expected values.	No claim that a candidate implements it.
Engineering conformance	Every applicable requirement passes on one exact candidate; no required skip, unresolved deviation, or false completion.	No claim that persisted WCS/response is scientifically faithful in real data.
Representation/response fidelity	Round-trip and injection evidence covers WCS, units, response classes, boundaries, and covariance representations with preregistered bounds.	No observational photometric, astrometric, or noise performance claim.
Observational performance	Independent references and repeatability establish the named amplitude, shape, position, covariance, or false-rate metric.	No production-readiness claim.
Production readiness	Operational profile, complete products, failure/log audit, dependency closure, release controls, and owner approval all pass.	No authority beyond the exact released profile and contract version.

An owner decision can block a claim without making the requirement disappear. Record the exact decision ID from the separate ledger and the conservative behavior observed.

4 Execution order and atomicity protocol

Run evidence in the following order so later evidence cannot mask an earlier contract failure:

1. Parse and freeze the exact frozen PTC r0.5 parent/application generation, transformed-signal and retention states, exact AST same- n coordinate parent, boundary/profile versions, and source digests. Reject direct CAL, PTC-disabled, missing-product, inferred no-op, and cross-generation routes.
2. Parse and freeze identity, units, lifecycle, MAP policy (including the exact admitted value and unit status of every policy parameter), WCS, and required-product plan.
3. Separately verify PTC-owned source facts, ALIGN/AST-owned coordinate facts, and MAP-owned projection/coefficient/boundary/support/use conditions before arithmetic. Verify that VAL, if used, evaluates the named rule without authoring it and that no layer rescues a failed condition in another.
4. Compare exact scientific output-row identity and cardinality, plus one-pixel, one-hot half-open-boundary, prohibited-fractional, covariance, response, and distinct-fact observation outputs, to independent expected values; retained storage rows do not count as scientific output rows.
5. Verify complete immutable observation-bundle construction.
6. Snapshot coadd-owned state; test every compatibility failure; require atomic nonmutation.
7. Exercise admitted centered-integer coaddition, exact coadd output-row identity, and companion placement.
8. Round-trip all required representations and reconstruct each row-selected operator from provenance.
9. Exercise honest response/covariance disclosure, including partial and unavailable states that leave the map valid, and verify versioned binding for later response, covariance, or corrected-map products.
10. Exercise consumer claim boundaries and immutable raw parentage without assuming a closed-world registry of all future scientific uses.
11. Repeat supported execution modes under the preregistered floating-point policy, while requiring exact discrete facts.

12. Only after deterministic gates pass, execute representation, external, observational, and operational evidence appropriate to the requested claim.

For every failure-injection test, capture pre-state, attempted input, emitted diagnostic, post-state, publication/cardinality state, and completion state. No partial-state comparison may omit a coadd companion merely because its payload was not the injected failure target.

5 Canonical requirement conformance clauses

The following clauses are generated from the shared authority. The first paragraph is the canonical science requirement; the second is its minimum conformance route. The result field intentionally remains unassessed.

SCI-MAP-REQ-001 – Versioned authority

Canonical clause: The scientific contract version shall be v0.1. The rendered document revision shall be recorded separately as r0.5; changing layout or exposition alone shall not be represented as a new scientific contract version.

Minimum conformance evidence: Inspect both artifacts and their machine-readable provenance for distinct contract-version and document-revision fields.

Result: UNASSESSED

SCI-MAP-REQ-002 – Capability and realized PTC input

Canonical clause: The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- x product $z_i = Z_i^{\text{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent mJy/beam convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

Minimum conformance evidence: Exercise exact realized PTC input, attempted CAL input, PTC-disabled and no-product routes, incompatible PTC generation, missing nominal-beam identity, and attempted Stokes-I promotion.

Result: UNASSESSED

SCI-MAP-REQ-003 – Exact occurrence identity

Canonical clause: Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

Minimum conformance evidence: Permute times, containers, rows, and columns while preserving or breaking the exact parent chain; products shall follow exact occurrence identity and reject inferred joins.

Result: UNASSESSED

SCI-MAP-REQ-004 – Producer facts and MAP admission

Canonical clause: SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/availability and coefficient-QC disposition without reinterpretation. It shall apply the occurrence-level profile SCI-MAP:map_upstream_admission01; VAL Registry binds and VAL Core evaluates that MAP-authored policy but does not author it. Every fact, decision, cause, and scope remains separately typed.

Minimum conformance evidence: Toggle each producer fact and MAP profile predicate independently; verify exact VAL evaluation, preserved causes, and no universal producer-flag action.

Result: UNASSESSED

SCI-MAP-REQ-005 – Coordinate and projection authority

Canonical clause: MAP shall consume SCI-AST:rtc_output_grid_coordinates01 for the same stable RTC sample n and complete parent chain as z_i ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through SCI-MAP:one_hot_containing_pixel01: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

Minimum conformance evidence: Test exact-parent joins, centers, lower and upper cell boundaries, outer extent, invalid coordinates, sum-over-pixel normalization on its in-grid domain, and prohibited wrap/clamp/fractional placement.

Result: UNASSESSED

SCI-MAP-REQ-006 – Ordinary coefficient status

Canonical clause: The ordinary route requires one exact PTC-owner-selected MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052–055 and the PTC-to-MAP boundary. No such family is selected in r0.5,

so ordinary numerical MAP production is unavailable until that owner decision is versioned and bound. MAP shall infer neither unity, loading, sensitivity, scatter, nor precision; later coefficient generations shall not alter earlier PTC products, and cross-generation use requires explicit compatibility.

Minimum conformance evidence: Attempt every inferred family and mismatched generation and require typed route unavailability; when a future family is bound, verify its complete identity, index/broadcast, statistic, factors, unit, normalization, population, support, lifecycle, use, QC, retention relation, uncertainty, and prohibited meanings.

Result: UNASSESSED

SCI-MAP-REQ-007 – Exposure lineage and accounting

Canonical clause: ALIGN-owned occurrence facts e_{acq} and e_{vo} , in seconds of physical native integration support, shall remain immutable through RTC/CAL/PTC. MAP shall derive distinct upstream-eligible, retained, one-hot-projected, and coadded exposure from deduplicated original-occurrence lineage under its exact profiles. Invalid-original has $e_{\text{vo}} = 0$; synthesized, surrogate, missing, or unoccupied support adds no acquired exposure; replacement and overlapping support create no new exposure. Boundary loss contributes nowhere. Exposure shall not be reconstructed from duration alone, cadence, sample count, hits, Q , γ , or formal precision.

Minimum conformance evidence: Use overlapping support, donor replacement, synthesis, invalid-original, and boundary-loss fixtures; compare stable-original-occurrence deduplicated seconds at every named stage.

Result: UNASSESSED

SCI-MAP-REQ-008 – Response information state

Canonical clause: MAP shall report the response information it actually provides, including meaning, domain, parentage, normalization, units, and limitations, or an explicit typed unavailable state. Missing or incomplete response shall not invalidate an otherwise valid map, fabricate an identity response, or prohibit later scientific analysis. A later simulated response or response-corrected map shall be a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Exercise complete, partial, and unavailable response states; verify honest claims, no sentinel or fabricated identity response, and versioned derivative-product binding.

Result: UNASSESSED

SCI-MAP-REQ-009 – Immutable plan and lifecycle

Canonical clause: The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

Minimum conformance evidence: Compare serialized states across two deliberately different observations and verify one-way provenance and complete replacement.

Result: UNASSESSED

SCI-MAP-REQ-010 – Factored contribution predicate

Canonical clause: For each pixel, membership in $\mathcal{C}_p$ shall require, as separately typed gates: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC disposition; MAP upstream admission; AST coordinate validity; finite z_i ; finite positive γ_i ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Unknown, conflict, ineligible, unavailable, invalid, and not-requested all prevent arithmetic where applicable but preserve distinct causes and scopes. Every gate shall be screened before numerical payload evaluation; no numeric-zero mask represents the predicate.

Minimum conformance evidence: Vary one gate and one knowledge state at a time, compare exact membership and cause/scope records, and place NaN/Inf behind excluded gates to prove pre-arithmetic screening.

Result: UNASSESSED

SCI-MAP-REQ-011 – Ordinary accumulators

Canonical clause: On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \gamma_i > 0$.

Minimum conformance evidence: Use one-pixel and one-hot hand calculations and compare accumulators before normalization; fractional projection is not authorized in v0.1.

Result: UNASSESSED

SCI-MAP-REQ-012 – Normalized publication

Canonical clause: The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\hat{m}_p = N_p/Q_p$, and an available response companion gives $\hat{k}_p = K_p/Q_p$. No unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero.

Minimum conformance evidence: Exercise positive, exact-zero, near-zero, overflowed, and non-finite aggregates; compare exact row identity and inspect any nonauthoritative storage slots.

Result: UNASSESSED

SCI-MAP-REQ-013 – Accumulator interpretation

Canonical clause: The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

Minimum conformance evidence: Audit schema, metadata, API types, and labels for every numerator and normalization representation.

Result: UNASSESSED

SCI-MAP-REQ-014 – Constant preservation

Canonical clause: With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

Minimum conformance evidence: Map constant fixtures with uniform and unequal coefficients; compare exact output-row membership, then confirm that stronger response claims remain separately gated.

Result: UNASSESSED

SCI-MAP-REQ-015 – Projection-specific estimand

Canonical clause: The authorized base projection is exactly `SCI-MAP:one_hot_containing_pixel@1`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

Minimum conformance evidence: Use containing-pixel and outer-boundary fixtures; attempt fractional or WLS substitution and require method rejection.

Result: UNASSESSED

SCI-MAP-REQ-016 – Same operator for linear companions

Canonical clause: Signal, available response/kernel tracer, and declared linear realizations shall use the same admitted membership, geometric placement, and exact support-authorized output-row selection; retained exposure and coadd observation count shall use the same membership and placement wherever their definitions require them.

Minimum conformance evidence: Trace coefficient-by-coefficient membership, placement, and scientific output-row selection through signal and each declared linear companion.

Result: UNASSESSED

SCI-MAP-REQ-017 – Realized response claim

Canonical clause: A stored kernel may be called the realized response to template τ only when its sample parent satisfies $k = H\tau$ and the support-restricted map product satisfies $\tilde{k}_{\text{out}} = A_{\text{out}}H\tau = R_{\text{out}}\tau$ on exactly the authorized rows. Otherwise it shall be labeled only a tracer or unavailable.

Minimum conformance evidence: Inject a declared template and an intentionally mismatched tracer; compare exact response-row membership and accept the response label only for the proven identity.

Result: UNASSESSED

SCI-MAP-REQ-018 – Absolute-offset mode

Canonical clause: SCI-MAP v0.1 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

Minimum conformance evidence: Use a constant-offset fixture and inspect both the mapped value and recorded upstream response/null-mode state.

Result: UNASSESSED

SCI-MAP-REQ-019 – Conditional covariance

Canonical clause: When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ , MAP shall propagate it as $C_{m,\text{out}} = A_{\text{out}} N_{\text{PTC}} A_{\text{out}}^T$, retaining consequential off-diagonal terms. Missing, limited, or unavailable terms remain typed and are never zero-filled; absence of a complete model does not invalidate the signal map.

Minimum conformance evidence: Compare a small exact propagation and honest summarized, incomplete, and unavailable states on identical row membership.

Result: UNASSESSED

SCI-MAP-REQ-020 – When normalization is precision

Canonical clause: Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

Minimum conformance evidence: Test one-hot geometry with both proven inverse-variance and nonprecision coefficient families.

Result: UNASSESSED

SCI-MAP-REQ-021 – Formal diagonal weight

Canonical clause: When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p , support-restricted precision $C_{m,\text{out}}^+$, exposure, hits, and empirical statistical weight.

Minimum conformance evidence: Compute and label every object for an authorized one-hot fixture and a correlated counterexample.

Result: UNASSESSED

SCI-MAP-REQ-022 – Covariance and nuisance disclosure

Canonical clause: The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Inspect every representation state, prevent promotion of normalization, hits, or diagnostic weight to covariance, and verify derivative-product binding without rewriting the original claim.

Result: UNASSESSED

SCI-MAP-REQ-023 – Data-derived state

Canonical clause: If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

Minimum conformance evidence: Use metadata cases for fixed and data-derived state and verify that unconditional labels appear only with separately admitted evidence.

Result: UNASSESSED

SCI-MAP-REQ-024 – No significance promotion

Canonical clause: Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not **sig2noise** or empirical significance. Empirical calibration remains NOI-owned.

Minimum conformance evidence: Attempt each prohibited relabeling through all public representations and require rejection or an unmistakable non-significance label.

Result: UNASSESSED

SCI-MAP-REQ-025 – Distinct map facts

Canonical clause: Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, upstream-eligible exposure, retained exposure, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

Minimum conformance evidence: Construct cases in which each pair differs and verify separately typed identities and immutable ownership.

Result: UNASSESSED

SCI-MAP-REQ-026 – Incidence layers

Canonical clause: AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in $\mathcal{C}_p$. The one-hot touch convention counts the unique containing pixel only.

Minimum conformance evidence: Use coordinate-valid/signal-unavailable, route-ineligible, zero-coefficient, and admitted occurrences and compare all three populations.

Result: UNASSESSED

SCI-MAP-REQ-027 – Estimator contributions

Canonical clause: Estimator-admitted contributions shall count exactly $i \in \mathcal{C}_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

Minimum conformance evidence: Compare exact counts with an independently evaluated profile and factored-gate oracle while preserving cause and decision scope.

Result: UNASSESSED

SCI-MAP-REQ-028 – Admitted observation count

Canonical clause: For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

Minimum conformance evidence: Use one complete, one unsupported, and one rejected observation bundle and verify the per-pixel count.

Result: UNASSESSED

SCI-MAP-REQ-029 – Upstream-eligible exposure

Canonical clause: MAP upstream-eligible exposure shall union and deduplicate the exact original occurrences carrying positive e_{vo} that influence MAP-route candidates admitted by `SCI-MAP:map_upstream_admission@1`, then place their seconds by the same one-hot containing-pixel rule. It precedes coefficient and MAP-local numerical rejection.

Minimum conformance evidence: Use eligible zero-coefficient and multiply influencing occurrences and verify deduplicated seconds without precision promotion.

Result: UNASSESSED

SCI-MAP-REQ-030 – Retained exposure

Canonical clause: MAP retained exposure shall union and deduplicate positive- e_{vo} original occurrences influencing estimator-admitted members and place them by the same one-hot rule. Coadded exposure shall union compatible observation-level original-occurrence sets after atomic coadd admission; repeated parents are not double counted. Neither quantity is precision.

Minimum conformance evidence: Use overlapping observation lineage and mixed admission and compare retained/coadded exposure with exact set unions and normalization.

Result: UNASSESSED

SCI-MAP-REQ-031 – Adopted threshold selector

Canonical clause: For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p ; for count $N > 0$, select zero-based $k = \lfloor ([0.75N] + N)/2 \rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population identity and exact dimensionless support-policy scalar $c = \text{coverage_cut}$ shall be recorded. The effective support policy shall explicitly admit that exact numerical c state before support resolution; otherwise the required product shall fail closed.

Minimum conformance evidence: Check empty and multiple population sizes against an independent integer-index oracle; inspect population identity and exact numerical c admission. Exercise negative, zero, non-finite, and greater-than-one c states without assuming their disposition; only an explicitly owner-authorized effective-policy state may proceed.

Result: UNASSESSED

SCI-MAP-REQ-032 – Separate support predicates

Canonical clause: Only after the effective support policy explicitly admits the exact value of the dimensionless scalar c , numerical normalization support shall require an explicit finite positive $Q_p \geq Q_\star c/10$, and science-policy support shall independently require an explicit finite positive $Q_p \geq Q_\star c$. This contract asserts no general numeric domain for c . Every non-authorized numerical state fails closed before a support row set is constructed. Neither support state shall establish final validity or precision.

Minimum conformance evidence: Exercise values below, at, and above both thresholds, including $N = 0$, and compare exact Boolean planes and output-row sets; verify fail-closed behavior for every numerical c state not explicitly admitted by the effective policy.

Result: UNASSESSED

SCI-MAP-REQ-033 – MAP-local base-product validity

Canonical clause: One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

Minimum conformance evidence: Use truth-table fixtures separating producer state, MAP admission, response/covariance disclosure, base validity, and later use-local eligibility.

Result: UNASSESSED

SCI-MAP-REQ-034 – Screen before arithmetic

Canonical clause: Invalid contributions shall be excluded before their numerical payload is evaluated. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

Minimum conformance evidence: Place NaN and Inf payloads behind false and true predicates and check contamination, diagnostics, and failure scope.

Result: UNASSESSED

SCI-MAP-REQ-035 – Exceptional contribution states

Canonical clause: Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

Minimum conformance evidence: Run one-cause-at-a-time fixtures and inspect cause-specific states plus unchanged neighboring pixels.

Result: UNASSESSED

SCI-MAP-REQ-036 – Complete observation bundle

Canonical clause: A base/unfiltered observation map shall be one immutable ordered bundle containing signal, numerator, normalization identity, response class/state, uncertainty/covariance state, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, lifecycle, exact PTC/AST parentage, and product identity. Limited or unavailable response/covariance states are complete disclosures, not automatic signal invalidity.

Minimum conformance evidence: Remove or alter each required identity or disclosure field and distinguish absent declarations from honest unavailable states.

Result: UNASSESSED

SCI-MAP-REQ-037 – Atomic coadd admission

Canonical clause: A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

Minimum conformance evidence: Snapshot all coadd-owned state before injecting one incompatibility at a time and require byte- or field-exact nonmutation.

Result: UNASSESSED

SCI-MAP-REQ-038 – Centered integer common grid

Canonical clause: Observation and coadd shapes shall differ by nonnegative even row and column counts, with the corresponding integer offset mapping their reference pixels to the same world coordinate. Fractional shift, reprojection, interpolation, wrapping, implicit recentering, and a different sky grid are forbidden.

Minimum conformance evidence: Test every parity/sign case and WCS reference-pixel relation with an independent coordinate oracle.

Result: UNASSESSED

SCI-MAP-REQ-039 – Ordinary coadd estimator

Canonical clause: The exact v0.1 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle admitted by `SCI-MAP:observation_coadd_admission@1`. It is an equal-observation coefficient, not precision or inverse variance. On exact centered-integer row identities, the coadd accumulates $N_p^c = \sum_o u_{op} \hat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and publishes $\hat{s}_p = N_p^c / Q_p^c$ only on $\mathcal{S}_{out}^c$.

Minimum conformance evidence: Use one and two observations, verify equal coefficients, exact row placement, output support, lifecycle identity, and rejection of precision labels.

Result: UNASSESSED

SCI-MAP-REQ-040 – Coadd companions and unavailable response

Canonical clause: The stacked signal vector $\widehat{\mathbf{m}}_{\text{obs}}$ shall carry explicit observation-row identities; B_{out} is the exact centered-integer equal-observation averaging operator; and T_{obs} is the stacked compatible observation-response object on those same domains. A numerical signal coadd may be produced under the named response-independent base-coadd role when any member response is unavailable, but the coadd response is then typed unavailable and no hidden response subset is formed. Exposure and count use the same atomic admission and placement.

Minimum conformance evidence: Use complete, limited, and unavailable member responses and verify unchanged signal membership, exact unavailable response, no zero companion, and no hidden subset.

Result: UNASSESSED

SCI-MAP-REQ-041 – Coadd covariance and precision

Canonical clause: C_{obs} shall be the block matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. For fixed admission, $C_{c,\text{out}} = B_{\text{out}}C_{\text{obs}}B_{\text{out}}^T$. Missing blocks are unknown, not zero or independence. The uniform observation coefficient is not precision; incomplete covariance limits only claims that require more.

Minimum conformance evidence: Use explicit within/cross-observation blocks plus missing-block cases and compare exact propagation and disclosure without invalidating the signal coadd.

Result: UNASSESSED

SCI-MAP-REQ-042 – Correlated-GLS boundary

Canonical clause: A correlated minimum-variance solution using $K_p^{-1}\mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

Minimum conformance evidence: Use positive- and negative-correlation covariance matrices that distinguish ordinary weighting from the GLS benchmark; require method identity to remain explicit.

Result: UNASSESSED

SCI-MAP-REQ-043 – Minimum product identity

Canonical clause: Every observation and coadd product shall identify its nonpolarimetric total-intensity-equivalent role; external observation/coadd, array/group, and occurrence domains; estimator and coefficient families; response/covariance states; signal unit and fixed-nominal-beam parent; complete AST frame/WCS; grid/projection/boundary; MAP profiles; exact PTC product/application and coefficient generations; exposure convention; lifecycle; parentage; causes; and product identity. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

Minimum conformance evidence: Round-trip every identity, detect collisions or generation mixing, and reject a bare token or serialization convention as scientific authority.

Result: UNASSESSED

SCI-MAP-REQ-044 – Frames and units

Canonical clause: MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are mJy/beam with exact nominal-beam identity; covariance has its square.

Minimum conformance evidence: Inspect complete frame and unit/beam identity and require cross-frame, ambiguous-frame, cross-unit, or cross-beam coadd rejection.

Result: UNASSESSED

SCI-MAP-REQ-045 – WCS and index persistence

Canonical clause: Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

Minimum conformance evidence: Round-trip representative centers, corners, axis signs, and index conversions; calculate maximum sky separation and exact orientation invariants.

Result: UNASSESSED

SCI-MAP-REQ-046 – Realized provenance

Canonical clause: Realized state shall record requested, effective, observation-resolved, applied, and realized identities; exact PTC/AST/boundary/profile generations; per-gate outcomes and causes; admitted contributions/bundles; accumulators and row domains; distinct map/exposure facts; PTC λ , $\hat{U}_{\text{corr}}$, $U_{\text{total},\Theta}$, removed-subspace and null-space identities; response class/state-change record; covariance state; offsets; required products; and failures.

Minimum conformance evidence: Reconstruct each applied row-selected operator, population, and typed unavailable state from provenance alone.

Result: UNASSESSED

SCI-MAP-REQ-047 – Failure scope and atomicity

Canonical clause: An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

Minimum conformance evidence: Inject each failure class at its boundary; inspect exact scope, diagnostics, nonmutation, and absence of a false completion marker.

Result: UNASSESSED

SCI-MAP-REQ-048 – Required observation products

Canonical clause: A complete observation bundle remains required when coaddition is enabled, and every product designated required by the effective v0.1 plan shall be published successfully before completion is asserted. Whether future abstract contracts require physical observation-map publication independent of an effective product policy remains an owner decision.

Minimum conformance evidence: Induce an observation-product publication failure during coadd operation and require overall failure with no misleading completion claim.

Result: UNASSESSED

SCI-MAP-REQ-049 – Immutable base parent

Canonical clause: Downstream transformations shall preserve resolvable base/unfiltered MAP-bundle identity and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, or sever parentage. Later response or covariance estimates are new versioned products.

Minimum conformance evidence: Create a later locally eligible product over a MAP-invalid parent and a later estimate over a MAP-valid parent; verify both states and original claims remain immutable.

Result: UNASSESSED

SCI-MAP-REQ-050 – Complete consumer boundary

Canonical clause: No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include quantity and nominal-beam identity, estimator/normalization/coefficient/profile identity, honest response class/state and uncertainty/covariance disclosure, distinct map and exposure facts, PTC additive-reference/removed-component/null-space state, exact frame/WCS, shape/indexing, lifecycle, parentage, dependency versions, and failures. Later analyses choose their own claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility.

Minimum conformance evidence: Exercise claims with missing required facts and analyses that explicitly accept limitations; verify fail-closed claim scope without rewriting MAP validity.

Result: UNASSESSED

SCI-MAP-REQ-051 – Same fixed noise operator

Canonical clause: An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

Minimum conformance evidence: Propagate fixed deterministic vectors through signal and realization paths and compare row identities and operator coefficients, not filenames or shapes.

Result: UNASSESSED

SCI-MAP-REQ-052 – Method and claim separation

Canonical clause: This authority shall not be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

Minimum conformance evidence: Attempt each method/claim promotion and verify explicit package/method rejection plus independent claim-state fields.

Result: UNASSESSED

6 Prediction execution matrix

Predictions are also imported unchanged. Engineering records attach fixtures, bounds, artifacts, and results; they do not rewrite the prediction.

SCI-MAP-PRED-001 – Uniform constant

Canonical prediction: For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized raw-valid output row; numerator and normalization scale together, and no unsupported scientific row is created.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-002 – Unequally weighted constant

Canonical prediction: For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-003 – One-pixel hand calculation

Canonical prediction: For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-004 – One-hot boundary

Canonical prediction: A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-005 – Delta input

Canonical prediction: A basis-pixel sky input produces the corresponding support-row-restricted column of $R_{\text{out}} = A_{\text{out}}H$, including projection and boundary truncation; it need not be a delta in the output map, and no value is predicted outside the authorized rows.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-006 – Point-source template

Canonical prediction: A declared point-source template τ_b produces $R_{\text{out}}\tau_b$ on exactly the support-authorized rows; peak, integrated, and fitted amplitudes are distinct functionals and need separate criteria.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-007 – Resolved and Fourier inputs

Canonical prediction: Resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{out}}\tau$ on the authorized row set; a constant-input pass does not predict preservation of these modes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-008 – Variable coverage

Canonical prediction: Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-009 – Finite boundary

Canonical prediction: At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-010 – Coefficient states

Canonical prediction: Until an exact PTC-owner-selected MAP-facing coefficient family is bound, the ordinary numerical route is unavailable. Under a future exact family, zero is noncontributing, negative/non-finite is invalid, and every state retains its cause; none is inferred as unity or precision.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states

Canonical prediction: A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or failing any required factored gate, does not contaminate an accumulator because exclusion occurs before payload evaluation. No universal flagged-candidate action is inferred.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-012 – Threshold selector

Canonical prediction: For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor ([0.75N] + N)/2 \rfloor$; empty input yields threshold zero, but no pixel gains support without its own finite positive coefficient. The dimensionless identity of c is fixed; any exact numerical c value not explicitly admitted by the owner-authorized effective policy fails closed before a support row set is constructed, and no numeric domain is inferred.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-013 – Missing response

Canonical prediction: A missing response parent yields a typed unavailable response state, never a fabricated zero or identity response. The original MAP bundle remains usable for claims that do not require response; any later simulated response or corrected map has a new versioned identity bound to that bundle.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-014 – Atomic rejection

Canonical prediction: A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-015 – One-observation coadd

Canonical prediction: A coadd containing one admitted observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. It reproduces response only when the exact response companion is available; otherwise response remains typed unavailable.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-016 – Uniform observation coefficients

Canonical prediction: On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. The normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-017 – Missing or invalid observation

Canonical prediction: An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-018 – Centered integer placement

Canonical prediction: Compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-019 – Cross-observation covariance

Canonical prediction: For two maps with variances σ_1^2, σ_2^2 and covariance c , the propagated uniform-coadd covariance includes c . A missing cross block remains unknown, never zero or independence, and the equal-observation normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-020 – Fixed noise operator

Canonical prediction: A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient A_{out} and B_{out} as the signal path; recomputed state exposes a named difference.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-021 – Sequential and parallel reduction

Canonical prediction: Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-022 – No false significance

Canonical prediction: A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-023 – No validity promotion

Canonical prediction: A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-024 – No ordinary predicate for JINC

Canonical prediction: A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.1.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

SCI-MAP-PRED-025 – Aggregate and index failure

Canonical prediction: An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion.

Primary claim layer: Engineering conformance.

Fixture / bound / result: TO BE PREREGISTERED

7 Mechanical completeness checks

Before a conformance verdict, automated checks shall establish:

- exactly one declaration of each stable requirement and prediction ID in the shared authority, with no gaps or duplicates;
- every requirement appears in this specification and in `CROSSWALK.md`; every prediction is linked to at least one requirement;
- each rendered PDF identifies contract v0.1 and its current document revision separately;
- no unresolved LaTeX reference, citation, overfull box, clipped glyph, broken table, or unreadable page remains;
- every required artifact reopens successfully, has the expected page count, and renders every page through an independent PDF renderer; and
- the owner-decision ledger and author-draft decision record contain no silent resolution of an open scientific question.

The conformance verdict is **pass** only when every applicable requirement has passing evidence on the exact candidate and all requested stronger claim layers have their own passing evidence. Otherwise the record is fail, blocked, or incomplete. This document itself records no verdict.

8 Required handoff contents

An engineering handoff contains the candidate and authority identities; requirement result table; prediction result table; exact commands and environment; independent expected-value artifacts; build/test/log outputs; representation and observational evidence where claimed; failure-injection snapshots; skipped/blocked items; deviations and owner dispositions; and an explicit statement for each claim layer. The handoff shall not collapse “conforms algebraically,” “represents faithfully,” “performs on sky,” and “ready for production” into one status word.